



**Ghulam Ishaq Khan Institute of Engineering Sciences
and Technology (GIKI)**

Project Scope Document

for

BlendGen: Blender Based Synthetic Dataset Generation for Object Detection

Version 1.0

by

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Acronyms

YOLO	You Only Look Once
COCO	Common Objects in Context
BVH	Bounding Volume Hierarchy

Abstract

Training modern object detection models requires large volumes of annotated data, which is often expensive, time consuming, and in many cases impractical to obtain. Real world data collection for specialized scenarios such as rare agricultural pests, controlled industrial environments, or edge case conditions introduces significant logistical and financial challenges. Existing dataset generation approaches either lack flexibility, require complex setups, or fail to provide sufficient control over object placement and environmental conditions. BlendGen addresses this bottleneck by introducing a Blender based addon for automated synthetic dataset generation with precise, surface constrained object placement. The system enables users to select arbitrary 3D surfaces, define object placement rules (including attachment, collision avoidance, and controlled randomness), and generate fully labeled image datasets in formats such as YOLO and COCO. By incorporating domain randomization techniques including variations in lighting, textures, camera perspectives, and environmental conditions. Blend Gen improves the generalization capability of models trained on synthetic data. To validate its effectiveness, datasets generated using BlendGen will be used to train object detection models (e.g., YOLOv8), with performance compared across synthetic only, real only, and hybrid training scenarios. The project aims to demonstrate that controlled synthetic data generation can significantly reduce dependency on manual data collection while maintaining practical performance in real world applications.

1. Introduction

The rapid advancement of computer vision has led to widespread adoption of object detection models across domains such as agriculture, surveillance, autonomous systems, and industrial automation. However, the effectiveness of these models is fundamentally dependent on the availability of large, well annotated datasets. Acquiring such datasets remains one of the most significant challenges in the machine learning pipeline.

Traditional data collection methods involve capturing real world images and manually annotating them, a process that is both labor intensive and expensive. In many cases, it is also infeasible particularly when dealing with rare objects, hazardous environments, or highly specific conditions that are difficult to reproduce consistently. As a result, researchers and developers often face a critical bottleneck that limits experimentation, slows development, and increases costs. Synthetic data generation has emerged as a promising alternative, leveraging 3D environments to create labeled datasets automatically. However, existing tools often require steep learning curves, lack intuitive workflows, or provide limited control over object placement and environmental constraints. This gap highlights the need for a system that is both powerful and accessible. BlendGen is proposed as a Blender native solution that simplifies and automates synthetic dataset generation while maintaining fine grained control over scene composition. By combining procedural generation techniques, geometric algorithms, and batch rendering within a familiar 3D environment, the system enables users to efficiently create high-quality labeled datasets for object detection tasks. While the system is general purpose, it can be effectively applied to scenarios such as agricultural pest detection (e.g., insects on plant leaves) or infrastructure monitoring (e.g., aircraft on runways), demonstrating its flexibility across domains.

2. Problem Statement

The development of robust object detection models is heavily constrained by the availability of large scale, high quality annotated datasets. In practice, collecting such data presents multiple challenges. Manual data collection and labeling require significant time, labor, and financial resources, making it difficult for students, researchers, and small teams to compete with larger organizations that have access to extensive datasets. Additionally, certain types of data are inherently difficult or impractical to obtain. Scenarios involving rare events, specific environmental conditions, or controlled object placements often cannot be captured reliably in the real world. For example, generating a dataset of pests consistently positioned on plant leaves under varying conditions, or aircraft appearing at specific angles on runways, would require substantial effort and coordination if done manually. Existing synthetic data generation tools provide partial solutions but often introduce their own limitations. Many systems are either overly complex, requiring significant expertise to configure, or lack precise control over object placement, collision handling, and environmental variability. This results in datasets that may not accurately reflect real world conditions or fail to generalize effectively when used for model training. As a result, there is a clear need for a system that enables efficient, controllable, and scalable dataset generation within an accessible environment. Such a system should allow users to define structured placement constraints, automate annotation generation, and introduce sufficient variability to support real world model performance without requiring extensive setup or specialized infrastructure.

3. Problem Solution/Objectives of the Proposed System

To address the limitations of traditional dataset collection and existing synthetic generation tools, BlendGen proposes a Blender based automated dataset generation system that combines procedural scene creation with precise control over object placement and environmental variability. The system enables users to define structured rules for placing objects onto selected 3D surfaces while ensuring realism through constraints such as surface adherence and collision avoidance. By automating both rendering and annotation, BlendGen eliminates the need for manual labeling and significantly accelerates dataset creation. In addition to controlled placement, the system incorporates domain randomization techniques to improve the robustness and generalization of models trained on synthetic data. These include variations in lighting, camera perspectives, object transformations, and environmental conditions. The generated datasets are exported in standard formats (e.g., YOLO, COCO), making them immediately usable for training modern object detection models. The effectiveness of the system will be validated through experiments involving model training on synthetic and hybrid datasets, demonstrating the practical viability of synthetic data in real world applications.

3.1 Objectives

BO-1: Develop a Blender based addon that automates synthetic dataset generation with minimal user intervention.

BO-2: Implement surface constrained object placement using geometric techniques (e.g., ray casting) to ensure realistic positioning of objects on arbitrary 3D meshes.

BO-3: Integrate collision detection mechanisms (e.g., BVH structures) to prevent unrealistic overlaps between objects during placement.

BO-4: Enable domain randomization by introducing controlled variations in lighting, textures, object transformations, and camera perspectives.

BO-5: Support generation of labeled datasets with automatic bounding box annotations in standard formats such as YOLO and COCO.

BO-6: Provide batch dataset generation capabilities with configurable parameters for scalability and experimentation.

BO-7: Design a reusable preset/template system to allow users to save and reuse dataset generation configurations.

- *BO-8: Validate the effectiveness of synthetic datasets by training object detection models (e.g., YOLOv8) and comparing performance across synthetic only, real only, and hybrid datasets.*
- *BO-9: Ensure the system remains accessible and user friendly within Blender, avoiding the complexity of existing high end synthetic data pipelines.*

4. Related System Analysis/Literature Review

Table 1 : Related System Analysis with proposed project solution

Application Name	Features	Weakness	Relevance with the Proposed System – How these weaknesses are addressed
BlenderProc	- Advanced synthetic data generation pipeline - Supports physics, lighting, and annotations	- Steep learning curve - Complex setup and scripting required	- BlendGen simplifies workflow through an intuitive Blender addon interface - Retains key functionality like controlled placement and annotation generation
NVIDIA Omniverse Replicator	- High fidelity rendering - Powerful domain randomization tools	- Requires high end hardware - Not accessible to students or small teams	BlendGen focuses on lightweight deployment within Blender without requiring specialized infrastructure
Unity Perception Toolkit	- Dataset generation with annotations - Integration with Unity engine	- Limited flexibility for arbitrary surface constraints - Requires Unity specific workflows	BlendGen enables precise surface-based placement directly on arbitrary meshes within Blender
Manual Data Collection	- Real world accuracy - No simulation gap	- Expensive and time consuming - Difficult for rare or specialized scenarios	BlendGen eliminates manual labeling and enables generation of rare or controlled scenarios on demand

5. Vision Statement

For students, researchers, and developers who face limitations in acquiring large scale annotated datasets, BlendGen is a flexible and accessible synthetic data generation tool that enables automated creation of high quality labeled datasets with precise control over object placement and environmental variability. Unlike traditional data collection methods, which are costly, slow, and often impractical for specialized scenarios, BlendGen provides a scalable and efficient alternative by leveraging procedural generation within Blender. The system empowers users to rapidly prototype datasets, experiment with different conditions, and train robust object detection models without being constrained by real world data availability.

6. Scope

BlendGen is a Blender based synthetic dataset generation system designed to streamline and automate the creation of annotated image datasets for object detection tasks. The system focuses on providing precise control over scene composition while maintaining ease of use and accessibility. At its core, BlendGen enables users to generate datasets by defining relationships between objects and surfaces within a 3D environment. Users can select target surfaces (e.g., plant leaves, ground planes, industrial components) and specify objects to be placed (e.g., pests, vehicles, tools). The system ensures realistic placement through surface constrained positioning and collision aware distribution, allowing multiple objects to be placed simultaneously without overlap. A key component of the system is its support for domain randomization, which introduces controlled variability in scene parameters. This includes changes in lighting conditions, camera angles, object transformations, and environmental properties, enabling the generation of diverse datasets that improve model generalization. The system will provide automated rendering and annotation generation, producing datasets in widely used formats such as YOLO and COCO. Batch processing capabilities will allow users to generate large datasets efficiently, while a preset/template system will enable reuse of configurations for different experiments.

Key features within the scope include:

- **Surface Constrained Object Placement:** Accurate placement of objects on arbitrary 3D surfaces using geometric techniques such as ray casting.
- **Collision Aware Multi Object Distribution:** Ensures realistic scene composition by preventing object overlap.
- **Domain Randomization:** Variability in lighting, textures, camera perspectives, and object properties to simulate diverse conditions.
- **Multi Camera Rendering:** Generation of images from multiple viewpoints to enrich dataset diversity.
- **Automated Annotation Generation:** Bounding box annotations in YOLO and COCO formats.
- **Batch Dataset Generation:** Scalable generation of large datasets with configurable parameters.

The system will be designed for Key features within scope include the above. Out of Scope: development of a full machine learning training platform, real world data collection or manual dataset labeling, support for multiple 3D engines (system is limited to Blender), pixel level annotation generation (e.g., segmentation masks), and cloud based deployment or distributed rendering infrastructure.

7. Project Stakeholders and Roles

Table 2: Project Stakeholders for BlendGen

Project Sponsor	GIKI
Stakeholder	• Abdul Hakeem, Syed Muhammad Faseeh, Zouhair Azam Khan • Muhammad Huzaifa Shah • Senior Design Project Evaluation Panel

8. References

- Artificial Intelligence Applications in Education. (2021, June 15). IEEE Conference Publication| IEEE Xplore. <https://ieeexplore.ieee.org/document/9477243>
- **Gupta, R. & Verma, P.** (2019, May 7). *Machine Learning and AI in Educational Systems*. Wiley. <https://www.wiley.com/enus/Machine+Learning+and+AI+in+Educational+Systems-p-9781119620025>
- **AI-Driven Academic Support Systems for Student Success.** (2025, March 10). *Element451 Blog*. <https://element451.com/blog/ai-driven-academic-support-systems>
- **Developing an Automated Student-Supervisor Matching System for Academic Projects.** (2024, August 15). *Dr. Shahabuddin Amerudin's Blog*. <https://people.utm.my/shahabuddin/?p=7417>
- **AI-Based Academic Advising Framework: A Knowledge Management Perspective.** (2023, August). *International Journal of Advanced Computer Science and Applications*, 13(8). <https://thesai.org/Publications/ViewPaper?Code=IJACSA&Issue=8&SerialNo=23&Volume=13>
- **Automated Similarity Detection: Identifying Duplicated Requirements.** (2019, December). *Universitat Oberta de Catalunya*. <https://openaccess.uoc.edu/bitstream/10609/105807/6/jmotgerTFM1219memory.pdf>
- **AI-DocGen - Automated Process Documentation Tool.** (2025, January 5). *GitHub Repository*. <https://github.com/CactusQuill/AI-DocGen>
- **AI in Higher Education: Enhancing Academic Achievement.** (2024, November 12). *eLearning Industry*. <https://elearningindustry.com/the-future-of-higher-education-integrating-ai-powered-learning-platforms-for-enhanced-academic-achievement>